

CSE 400: Fundamentals of Probability in Computing

Milestone 4 Scribe

Question-1

What is the probabilistic problem being addressed in your project? Clearly state the system objective and identify the primary sources of uncertainty involved.

Solution:

The probabilistic problem addressed in this milestone is the efficient and stable propagation of uncertainty in glacier melt modeling using an improved randomized algorithm.

In Milestone 3, uncertainty was propagated using Monte Carlo (MC) simulation, where independent random samples were drawn for each climatic variable. However, this approach resulted in high variance and incomplete exploration of extreme scenarios due to random clustering.

The objective of this milestone is to replace naive randomness with a structured sampling strategy using Latin Hypercube Sampling (LHS), ensuring systematic coverage of the probability space.

Primary sources of uncertainty remain:

- Air Temperature (T_a)
- Precipitation (P)
- Ice Albedo (α)
- Cloudiness (n)
- Snowfall Threshold (T_s)

These variables are propagated through the physical melt model:

$$Q_m = (1 - \alpha)S_0 + C_{sensible}T_a$$

$$M = \frac{Q_m \Delta t}{\rho_w L_f}$$

Thus, the problem is framed as a **multi-variable uncertainty propagation system with improved sampling efficiency**.

Question-2

Identify the key random variables and describe how uncertainty is modeled.

Solution: The same five environmental variables are modeled with fixed distributions:

- **Air Temperature:** $T_a \sim \mathcal{N}(0, 0.433^2)$, bounded using Chebyshev:

$$T_a \in [-1.3, 1.3]$$

- **Precipitation:** $P \sim \text{Gamma}(k = 2.0, \theta = 1.25)$, $P \geq 0$

- **Ice Albedo:** $\alpha \sim \mathcal{N}(0.35, 0.075^2)$, clipped:

$$\alpha \in [0.2, 0.5]$$

- **Cloudiness:** $n \sim \mathcal{N}(0.5, 0.15^2)$, clipped:

$$n \in [0, 1]$$

- **Snowfall Threshold:** $T_s \sim \mathcal{N}(1.0, 0.5^2)$

$$T_s \in [-0.5, 2.5]$$

Key Change: Instead of independent sampling (MC), uncertainty is now generated using **Latin Hypercube Sampling (LHS)**.

Probabilistic Assumptions:

- Variables are independent
 - Distributions remain unchanged from M3
 - Sampling is stratified across equal-probability intervals
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Question-3

Explain how probabilistic relationships are used within your system.

Solution: The system maintains independence at the input level but introduces **structured dependency through the LHS sampling matrix**.

In LHS:

- Each variable is divided into N equal-probability intervals
- One sample is drawn from each interval
- Samples are randomly paired to form N input vectors

Thus, while variables remain statistically independent, they are **systematically combined**, ensuring complete coverage of the joint probability space.

Dependencies arise through the physical model:

- T_a influences melt energy
 - α affects absorbed radiation
 - n controls transmissivity
 - P contributes to melt or accumulation
 - T_s determines phase transition
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Question-4

Describe how the probabilistic model aligns with implementation.

Solution: The implementation pipeline consists of:

- Generation of a $N \times 5$ LHS matrix in $[0, 1]$
- Transformation using inverse CDF (PPF):
 - Normal $\rightarrow$ `norm.ppf()`
 - Gamma $\rightarrow$ `gamma.ppf()`
- Application of physical bounds using clipping
- Propagation through the 3-stage melt model

The physics remains unchanged:

$$\tau = 0.6(1 - 0.5n)$$

$$Q_m = S_0\tau(1 - \alpha) + C_{sensible}T_a$$

$$M = \frac{Q_m \Delta t}{\rho_w L_f}$$

Key assumption:

- Only sampling method changes (MC $\rightarrow$ LHS)
 - Physical model remains constant
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Question-5

Describe the current state of the probabilistic model.

Solution: Well-defined components:

- All 5 random variables and distributions
- Physical melt equations

- LHS sampling mechanism
- Fixed parameter values and bounds

Key Achievements:

- Implementation of Latin Hypercube Sampling
- Reduced variance compared to Monte Carlo
- Stable mean estimates at $N = 5000$

Expected refinements:

- Correlation between variables
- Improved physical modeling
- Validation using real-world data

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Question-6

Identify unresolved probabilistic questions and responsibilities.

Solution:Unresolved Questions:

- How much variance reduction does LHS achieve compared to MC?
- Which variable contributes most to uncertainty?
- How reliable is the 95th percentile risk estimate?

Responsibilities:

- Variance Analysis Lead → MC vs LHS comparison
- Sensitivity Lead → correlation analysis
- Risk Analysis Lead → extreme case extraction